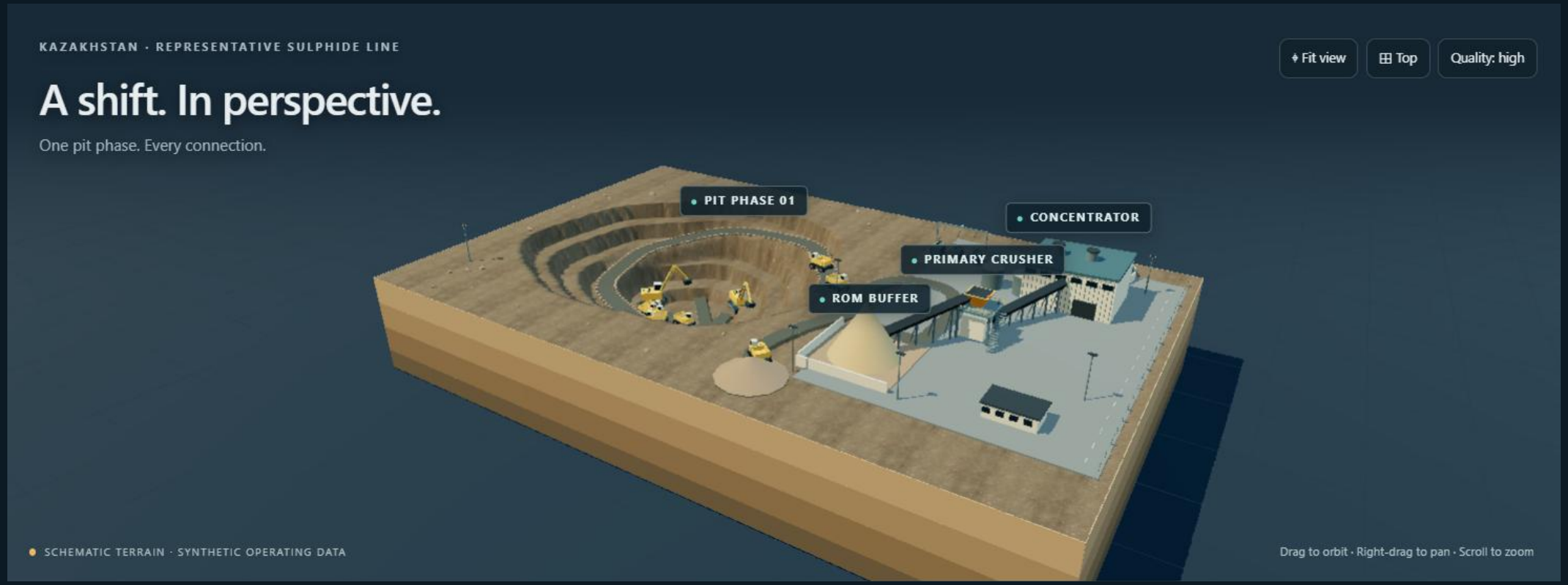
AKTOGAY

Sulphide Shift Twin



Tau-Ken Labs (Team-064)

Majhabul Hoque Tahsim • Sole member & developer

A real mine. A focused model.

Aktogay, Kazakhstan

Real-world reference

Open-pit copper operation with sulphide processing. The project represents one production line.

Model boundary

Pit faces, haulage, ROM stockpile, primary crusher and concentrator feed.

25 Mt/year

Representative line design basis

Synthetic operating data
Schematic terrain
No site calibration

The shift decision

How can the dispatcher keep plant feed steady after a disruption?

Dispatcher

Reassign trucks as equipment and haul routes change.

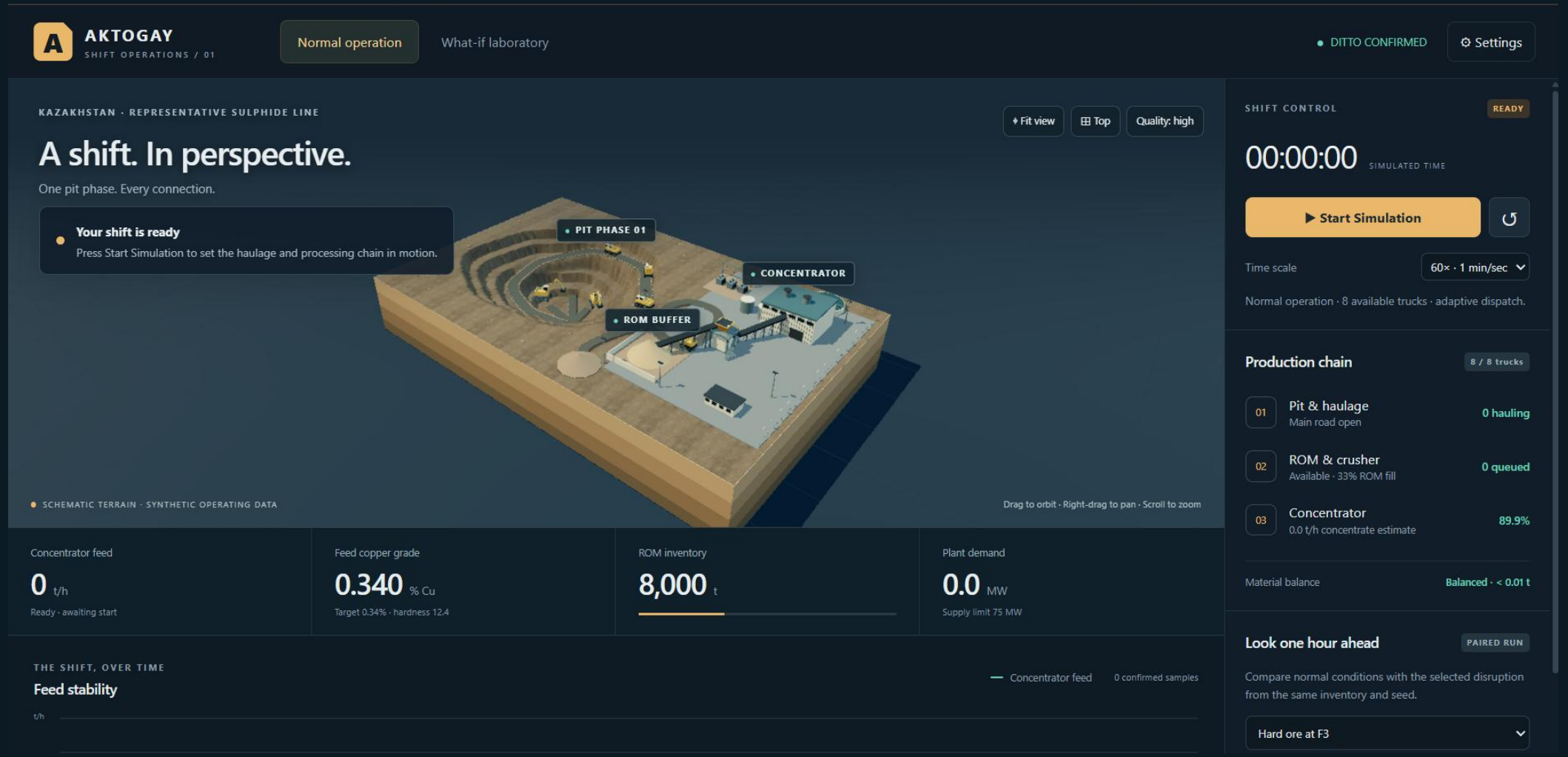
Mining engineer

Balance copper grade and ore hardness across available faces.

Plant supervisor

Track incoming tonnes, inventory and power constraints.

The operator workspace



One view connects equipment, shift controls and production KPIs.

Normal operation in motion



3,145

t/h concentrator feed

0.341%

Cu feed grade

66.0 MW

Synthetic snapshot

A controllable shift

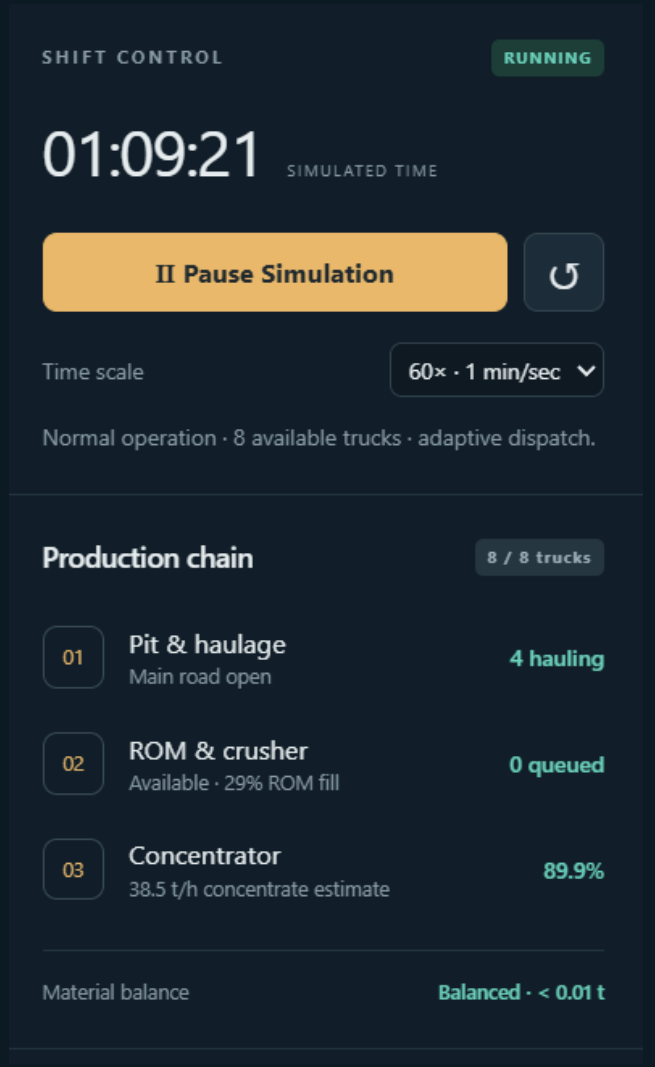
Start, pause and reset

Run the model deliberately, pause to inspect an event, or begin a fresh shift.

Adjust the pace

The captured run uses 60× speed: one simulated minute per second.

Visible material balance: < 0.01 t



The original six scenarios

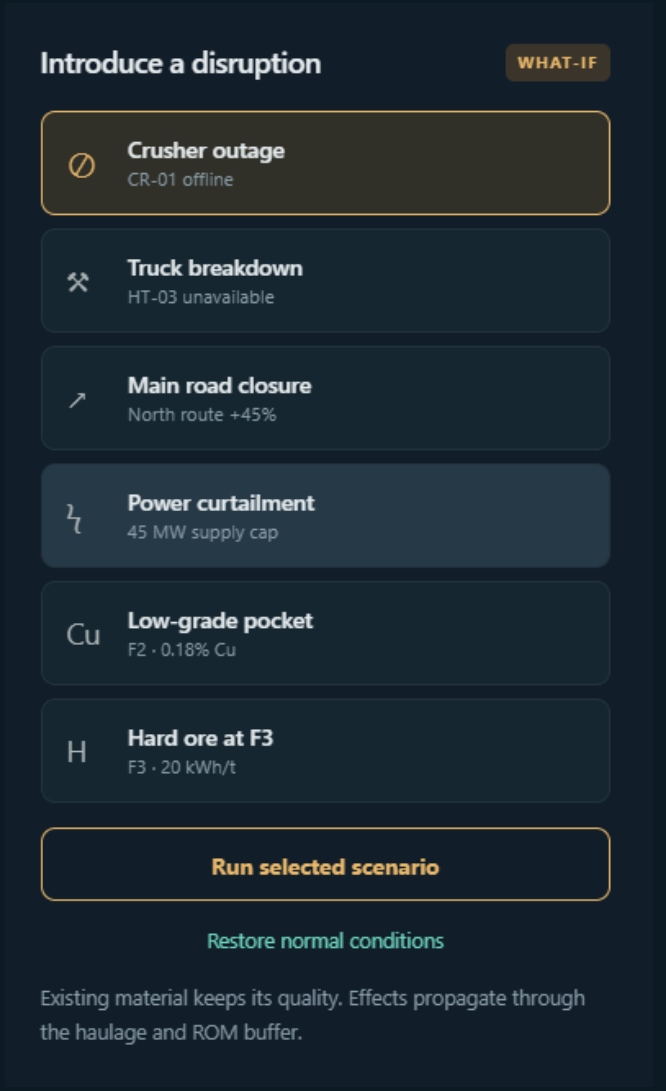
Equipment and routes

Crusher outage, truck breakdown and main road closure change flow and cycle time.

Power and ore quality

Power curtailment, low-grade F2 and hard F3 expose energy and blend constraints.

Select a scenario, run it, then restore normal conditions.



Crusher outage stops plant feed



0 t/h feed

8,317 t ROM inventory

Fault visible in the trend

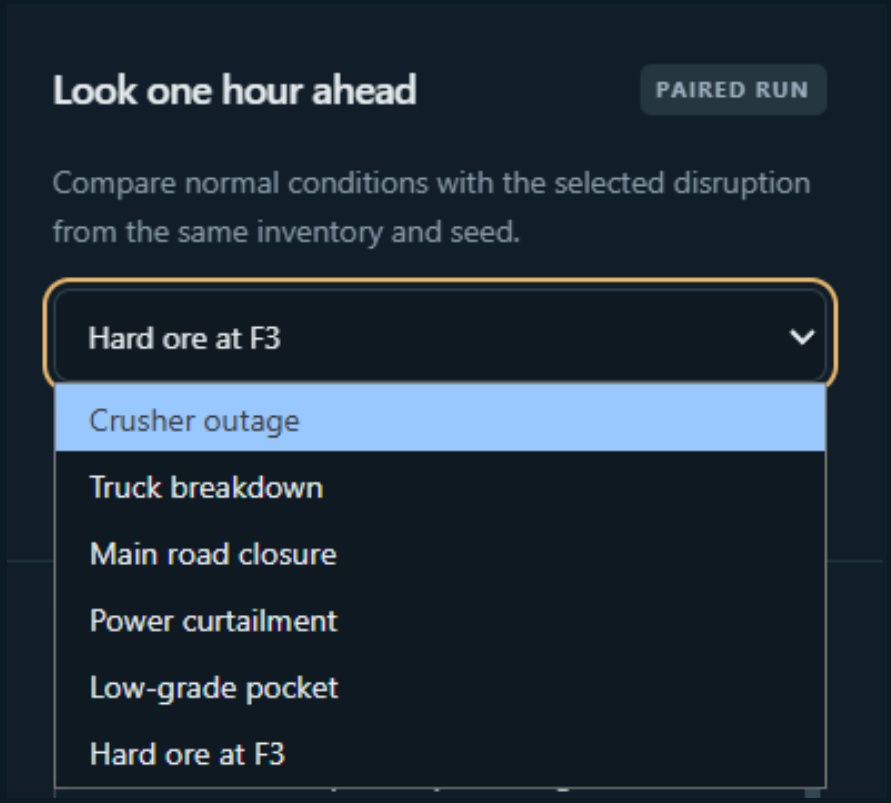
A one-hour decision experiment

Same starting state. Two possible shifts.

Baseline versus disruption

Compare normal conditions with the selected fault over the next 60 simulated minutes.

Inventory and random seed stay matched.



Settings change the actual model

Operating capacity

Fleet size, crusher throughput and power supply set the available resources.

Material and timing

Ore grade, hardness and cycle assumptions shape the downstream response.

Pause before applying a new shift setup.



3D exploration and graphics

The spatial view follows the simulation state



Inspection

Orbit, pan, zoom and select equipment to understand its current state.

Rendering choices

Fit or top view, selectable quality and locally bundled graphics assets.

One authoritative operator snapshot

01 Simulation model

Truck cycles, queues, ROM mixing, processing and energy constraints.

02 MQTT and Eclipse Ditto

Publish state and read the confirmed operator snapshot back.

03 3D operator interface

Clock, vehicles, KPIs and events share the same confirmed state.

04 Telemetry history

Telegraf, InfluxDB and Grafana support the platform’s data pipeline.

Freshness checks identify stale data instead of presenting it as live.

Evidence and engineering boundaries

31

model/API tests passed

25

matched policy comparisons

Evidence in this deck

Developer screenshots show the previous working version. New tests cover v3 constraints.

Remaining boundary

V3 browser and Docker integration remain unverified here. Operating inputs are synthetic.

Finite resources constrain the shift

Reserves are consumed

Loaded ore leaves the face reserve. Depleted faces stop supplying trucks.

Roads admit a limited flow

Distance and gradient affect travel. Entry headway creates queues.

Shovels hold real payload

Bucket passes set service time. An outage holds a partial load until recovery.

Material stays accounted for

Waste and tailings have their own twins. Five residuals check material flow.

Two additional disruption scenarios

SH-02 outage

F2 loading stops. Partial payload stays in place. New assignments use available faces.

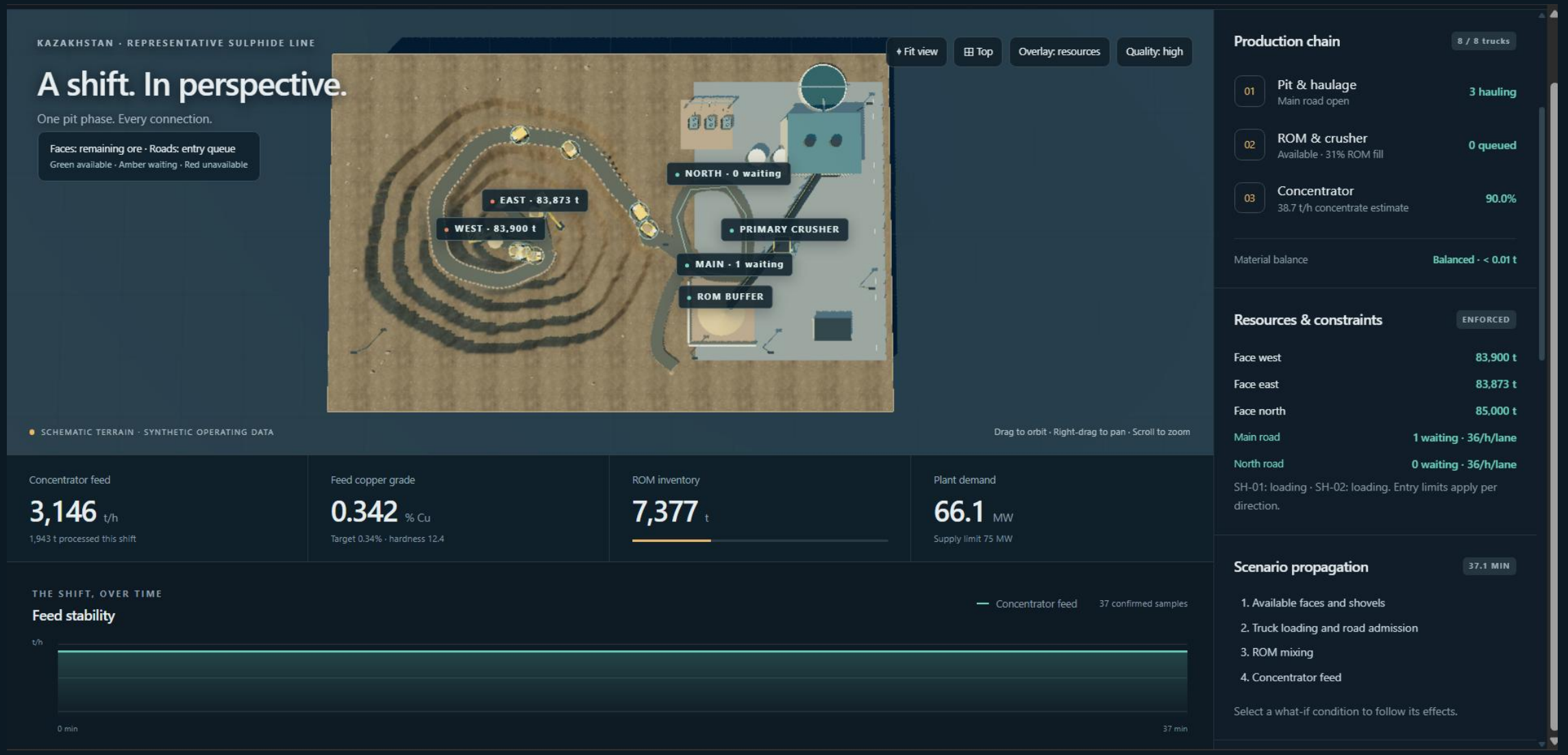
Road entry restriction

Main-road capacity falls to 12 vehicles per hour per direction. Queues build before entry.

Resource overlay

Remaining reserves, road queues and faults connect the spatial view to the model state.

Resource overlay in the running model



Dispatch policy evidence

Baseline: +237 t processed

Mean across five matched seeds over four simulated hours.

A visible tradeoff

Low-grade condition: +222 t processed, with grade error worse by 0.0051 percentage points.

Inventory matters

Compare final ROM too. Some gains consume the existing buffer rather than increase supply.

Pause the shift, choose a condition, compare fixed rotation and adaptive dispatch.

Judge demonstration

01 Start baseline and enable the resource overlay

02 Apply congestion, then a shovel outage

03 Pause and compare both dispatch policies

04 Inspect low-grade tradeoffs and material balances

05 Export the confirmed report

Tau-Ken Labs

A working shift twin for Aktogay’s production chain

Live demo: normal operation, disruption, recovery

Majhabul Hoque Tahsim

Sole member & developer / Team-064